HealthGuard ML Report

This automated report summarizes the machine learning training outputs for HealthGuard.

Included sections: dataset summary, model comparison, confusion matrices, ROC curves, clustering results, feature importance analysis and best model summary.

Dataset Summary

Metric	Value
Raw rows	1025.0000
Rows after duplicate removal	302.0000
Columns	14.0000
Feature columns	13.0000
Missing values	0.0000
Duplicate rows removed	723.0000
Heart disease cases	164.0000
No heart disease cases	138.0000

Dataset Notes

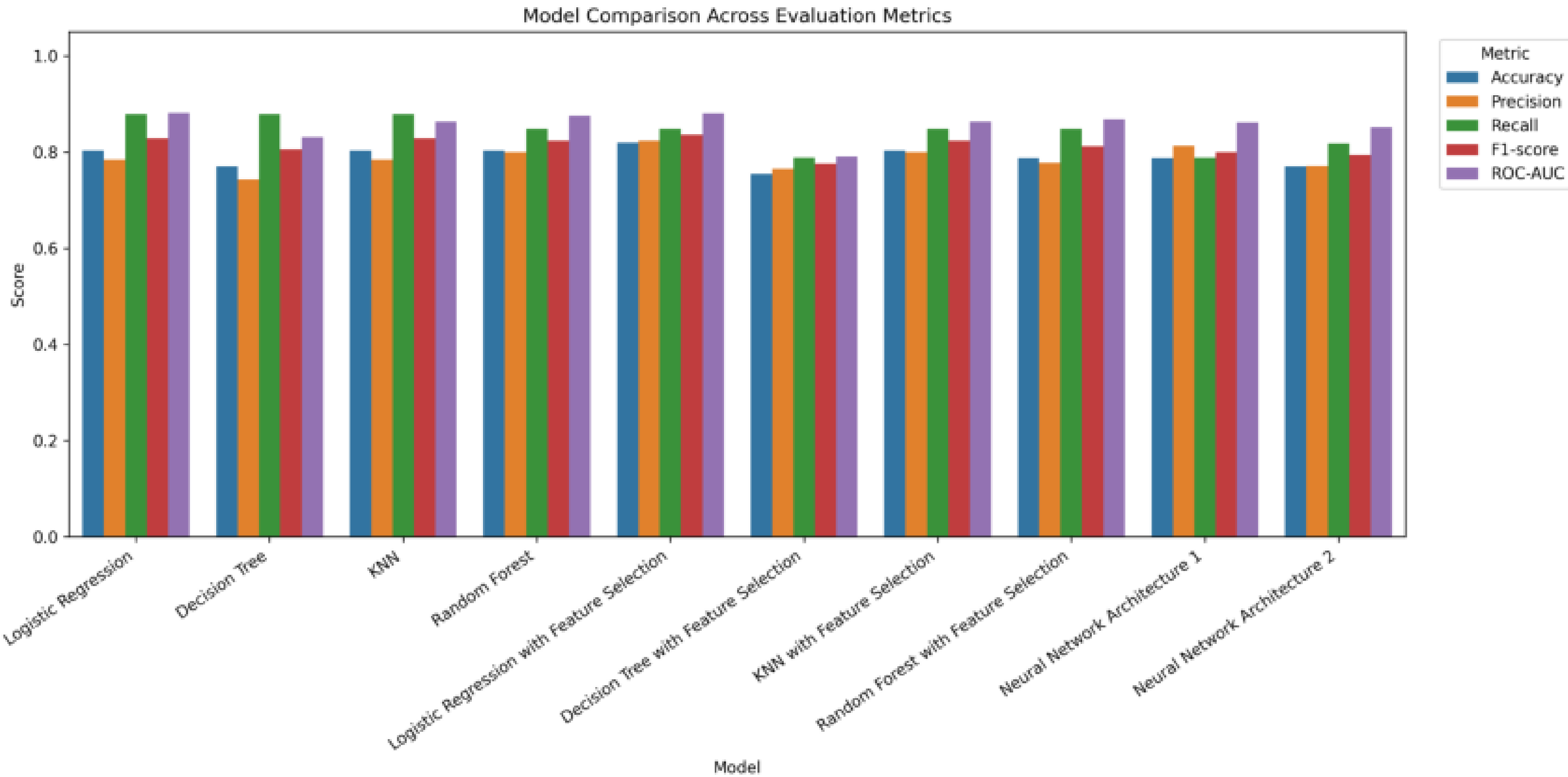
The report is generated from the Heart Disease dataset used by the HealthGuard machine learning notebook. Duplicate rows are removed in the notebook before model training, so the cleaned dataset counts are shown here.

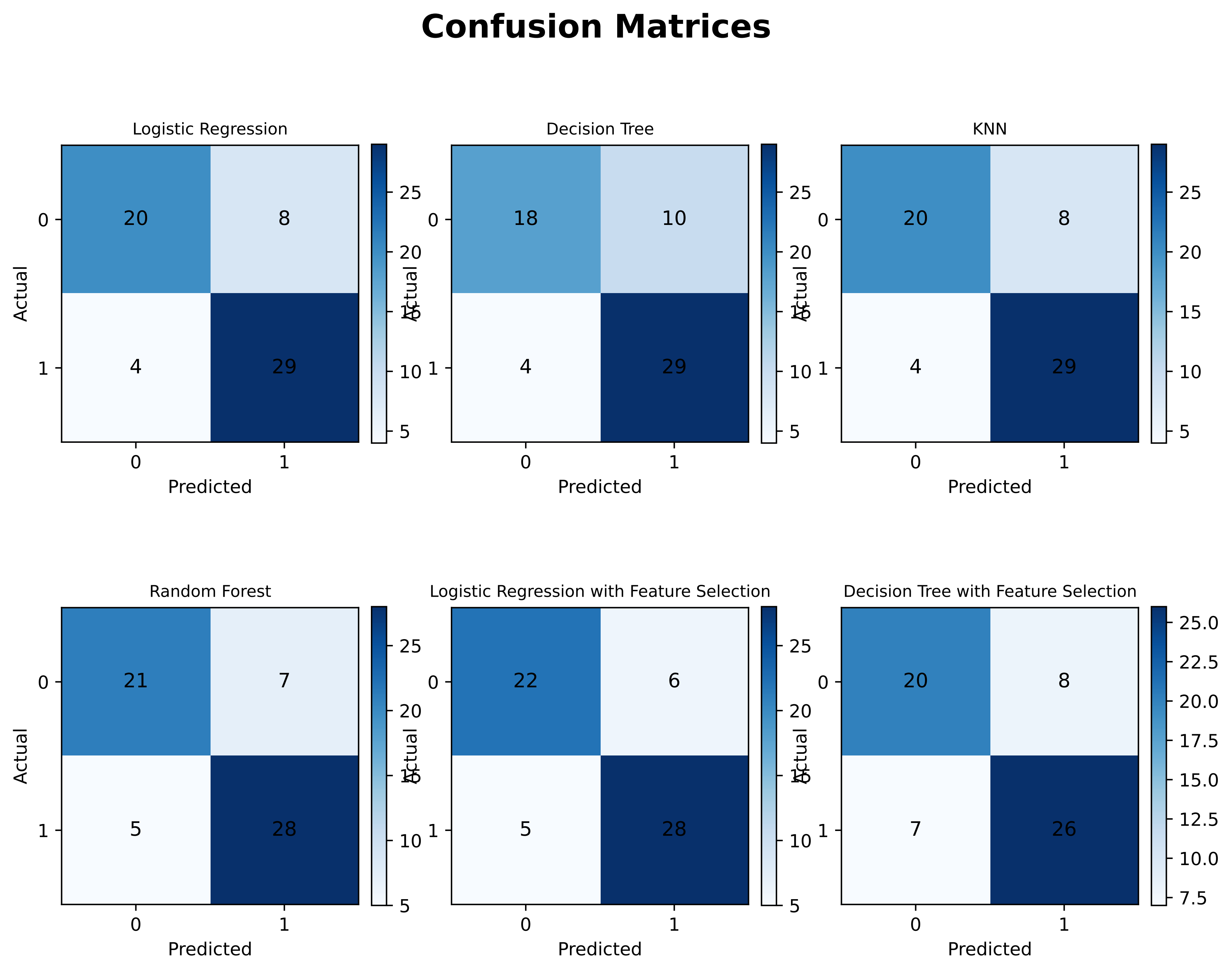
The target variable uses 1 for heart disease and 0 for no heart disease. All model, clustering and feature importance summaries in this PDF are generated from saved training outputs in ml/results.

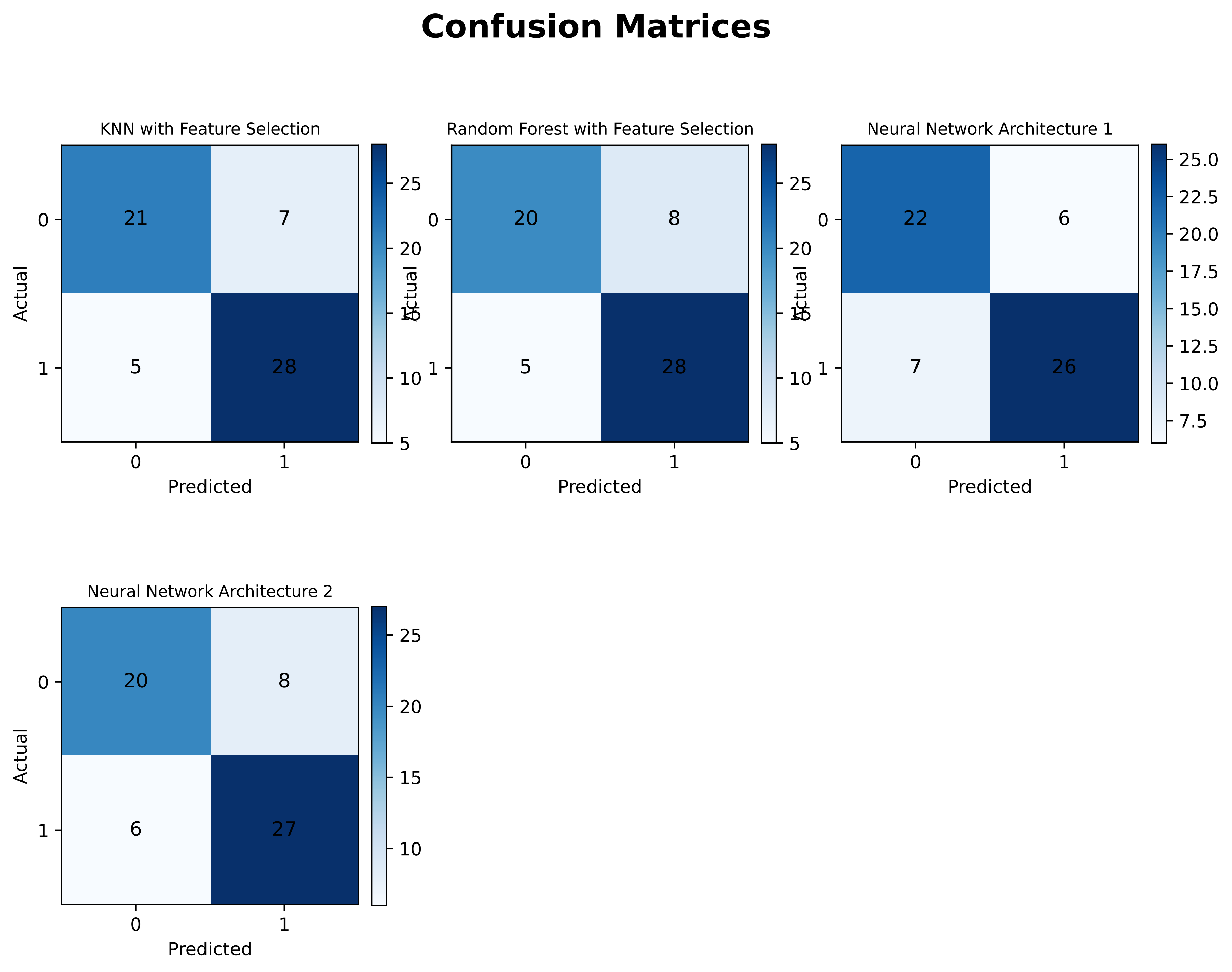
Model Comparison Table

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.8033	0.7838	0.8788	0.8286	0.8810
Decision Tree	0.7705	0.7436	0.8788	0.8056	0.8306
KNN	0.8033	0.7838	0.8788	0.8286	0.8626
Random Forest	0.8033	0.8000	0.8485	0.8235	0.8755
stic Regression with Feature Se	0.8197	0.8235	0.8485	0.8358	0.8799
ecision Tree with Feature Selec	0.7541	0.7647	0.7879	0.7761	0.7900
KNN with Feature Selection	0.8033	0.8000	0.8485	0.8235	0.8626
ndom Forest with Feature Selec	0.7869	0.7778	0.8485	0.8116	0.8680
Neural Network Architecture 1	0.7869	0.8125	0.7879	0.8000	0.8615
Neural Network Architecture 2	0.7705	0.7714	0.8182	0.7941	0.8517

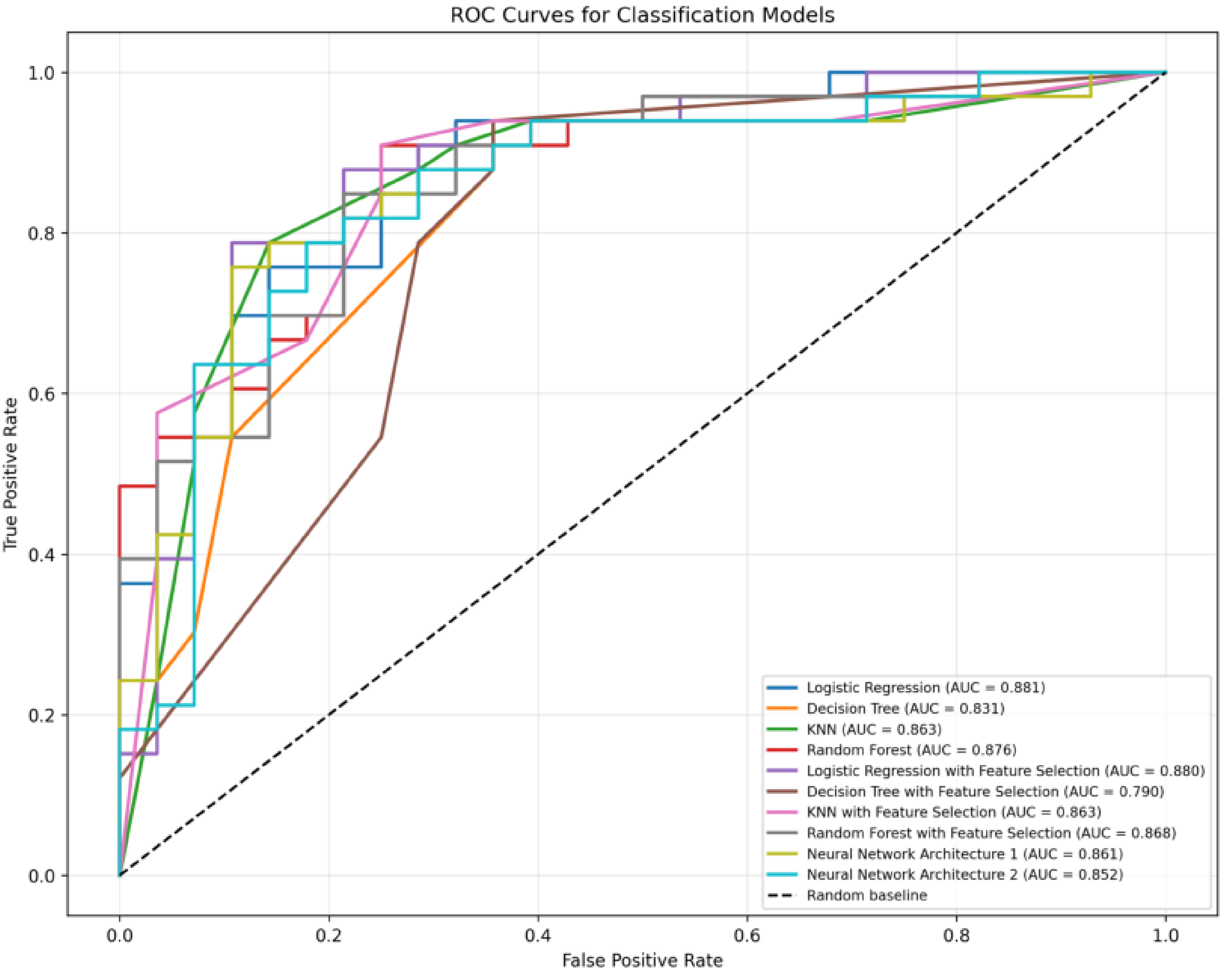
Model Comparison Chart




Confusion Matrices
Logistic Regression
Decision Tree
KNN
Random Forest
Logistic Regression with Feature Selection
Decision Tree with Feature Selection
Actual
Predicted
20
8
4
29
18
10
4
29
20
8
4
29
21
7
5
28
22
6
5
28
20
8
7
26


Confusion Matrices
KNN with Feature Selection
Random Forest with Feature Selection
Neural Network Architecture 1
Neural Network Architecture 2
Actual
Predicted
0
1
21
7
5
28
20
8
5
28
22
6
7
26
20
8
6
27

ROC Curves


ROC Curves for Classification Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.881)
Decision Tree (AUC = 0.831)
KNN (AUC = 0.863)
Random Forest (AUC = 0.876)
Logistic Regression with Feature Selection (AUC = 0.880)
Decision Tree with Feature Selection (AUC = 0.790)
KNN with Feature Selection (AUC = 0.863)
Random Forest with Feature Selection (AUC = 0.868)
Neural Network Architecture 1 (AUC = 0.861)
Neural Network Architecture 2 (AUC = 0.852)
Random baseline

Best Model Summary

Best model by F1-score: Logistic Regression with Feature Selection.

Key metrics: Accuracy=0.8197, Precision=0.8235, Recall=0.8485, F1-score=0.8358, ROC-AUC=0.8799.

F1-score is used as the primary summary metric because it balances precision and recall, which is important when heart disease screening must manage both false positives and false negatives.

Cross-Validation Stability

Model	Accuracy Mean	Accuracy Std	Precision Mean	Precision Std	Recall Mean	Recall Std	F1 Score Mean	F1 Score Std
KNN with Feature Selecti	0.8543	0.0223	0.8422	0.0505	0.9087	0.0505	0.8716	0.0162
om Forest with Feature S	0.8479	0.0391	0.8307	0.0529	0.9146	0.0887	0.8661	0.0371
: Regression with Feature	0.8477	0.0321	0.8362	0.0438	0.9025	0.0799	0.8644	0.0319
Logistic Regression	0.8279	0.0450	0.7953	0.0527	0.9267	0.0626	0.8539	0.0390
eural Network Architectu	0.8378	0.0432	0.8404	0.0424	0.8718	0.1024	0.8512	0.0484
Random Forest	0.8279	0.0368	0.8090	0.0496	0.9023	0.0678	0.8502	0.0325
eural Network Architectu	0.8213	0.0336	0.8347	0.0577	0.8475	0.0836	0.8361	0.0352
KNN	0.8046	0.0263	0.7937	0.0198	0.8657	0.0597	0.8270	0.0286
sion Tree with Feature Se	0.7713	0.0635	0.7487	0.0587	0.8845	0.0749	0.8082	0.0468
Decision Tree	0.7614	0.0615	0.7449	0.0558	0.8663	0.0908	0.7973	0.0483

Model Stability Notes

Cross-validation reports mean and standard deviation across five stratified folds. Higher mean values indicate stronger average performance; lower standard deviations indicate more stable behavior across different patient subsets.

Models with similar F1-score means should be compared by their F1-score standard deviation. A lower standard deviation suggests the model is less dependent on one favorable train-test split.

Clustering Results

k	Inertia	Silhouette Score	Adjusted Rand Index	Normalized Mutual Informatio
2.0000	2652.9010	0.1683	0.4110	0.3502
3.0000	2438.2792	0.1171	0.2558	0.2595
4.0000	2266.0572	0.1209	0.1826	0.1902
5.0000	2153.9783	0.1187	0.1626	0.1882
6.0000	2065.4597	0.1120	0.1299	0.1879
7.0000	2000.6129	0.1261	0.1500	0.1943
8.0000	1935.8368	0.1100	0.1118	0.1905
9.0000	1877.0214	0.1041	0.1206	0.2021
10.0000	1811.7163	0.0998	0.1188	0.2114

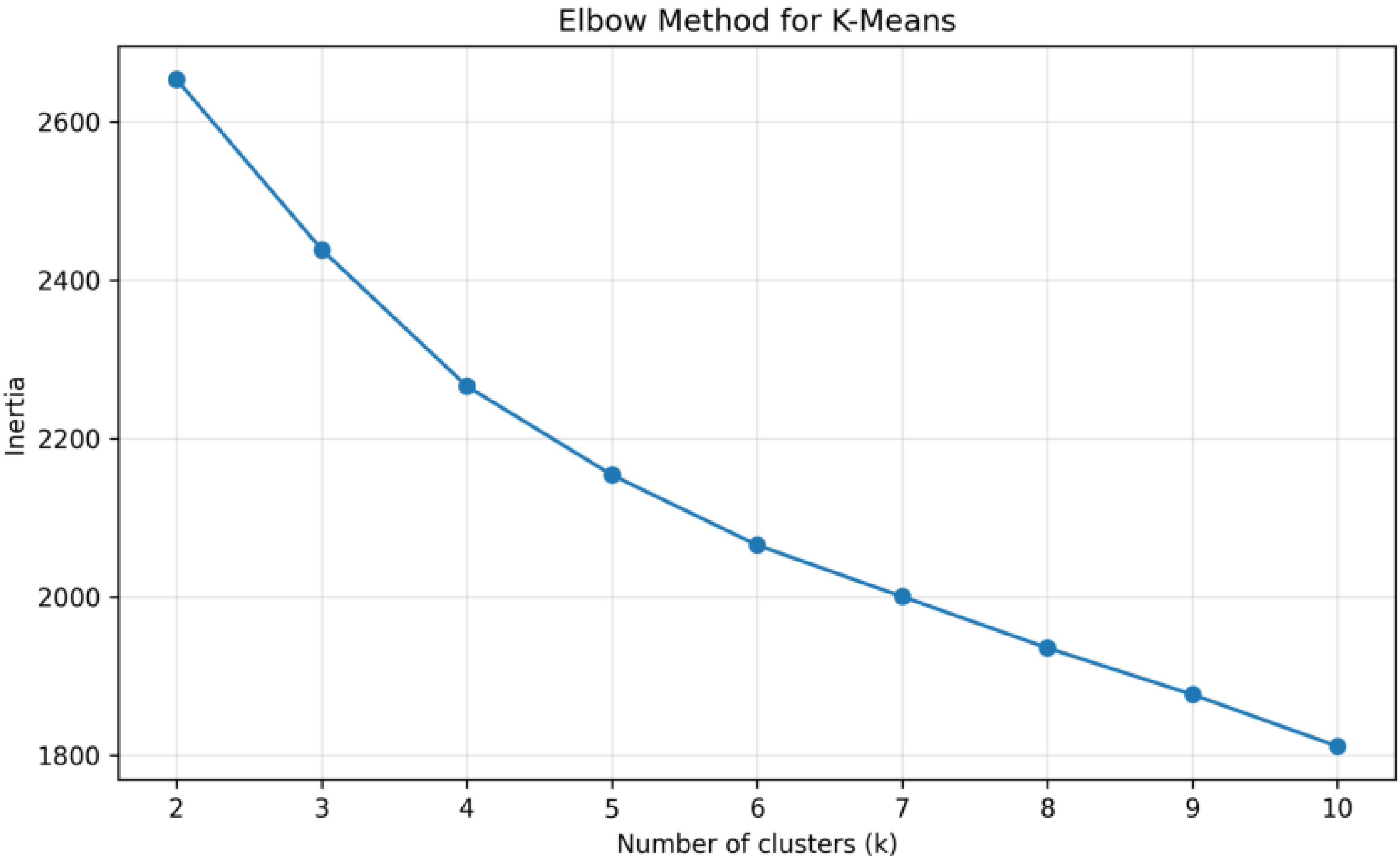
Clustering Summary

The best cluster count selected by Silhouette Score was k=2.

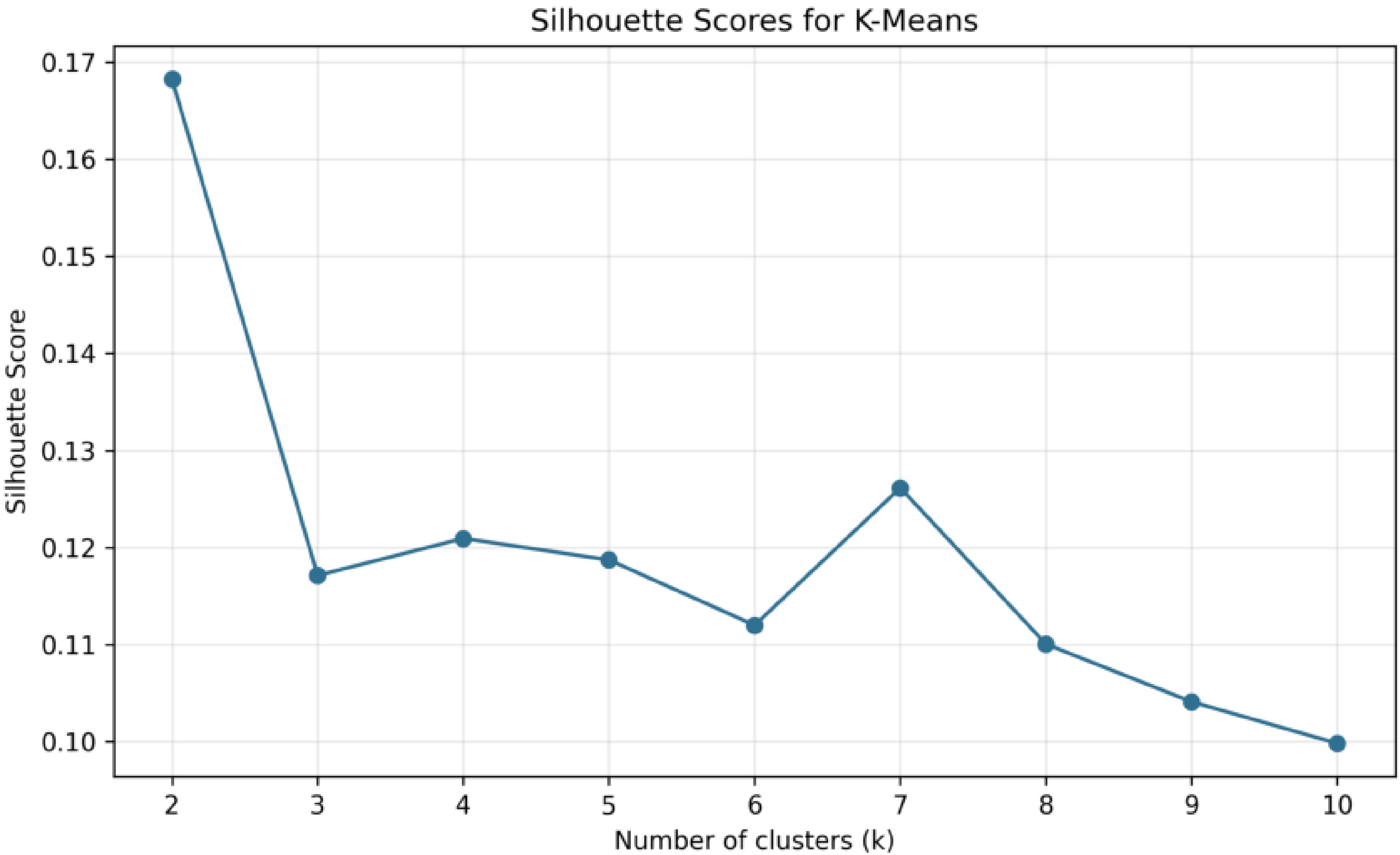
For this k, Silhouette Score=0.1683, Adjusted Rand Index=0.4110, and Normalized Mutual Information=0.3502.

ARI and NMI compare unsupervised cluster assignments with the known heart disease labels. These metrics are diagnostic only; the labels are not used by K-Means during clustering.

Elbow Method



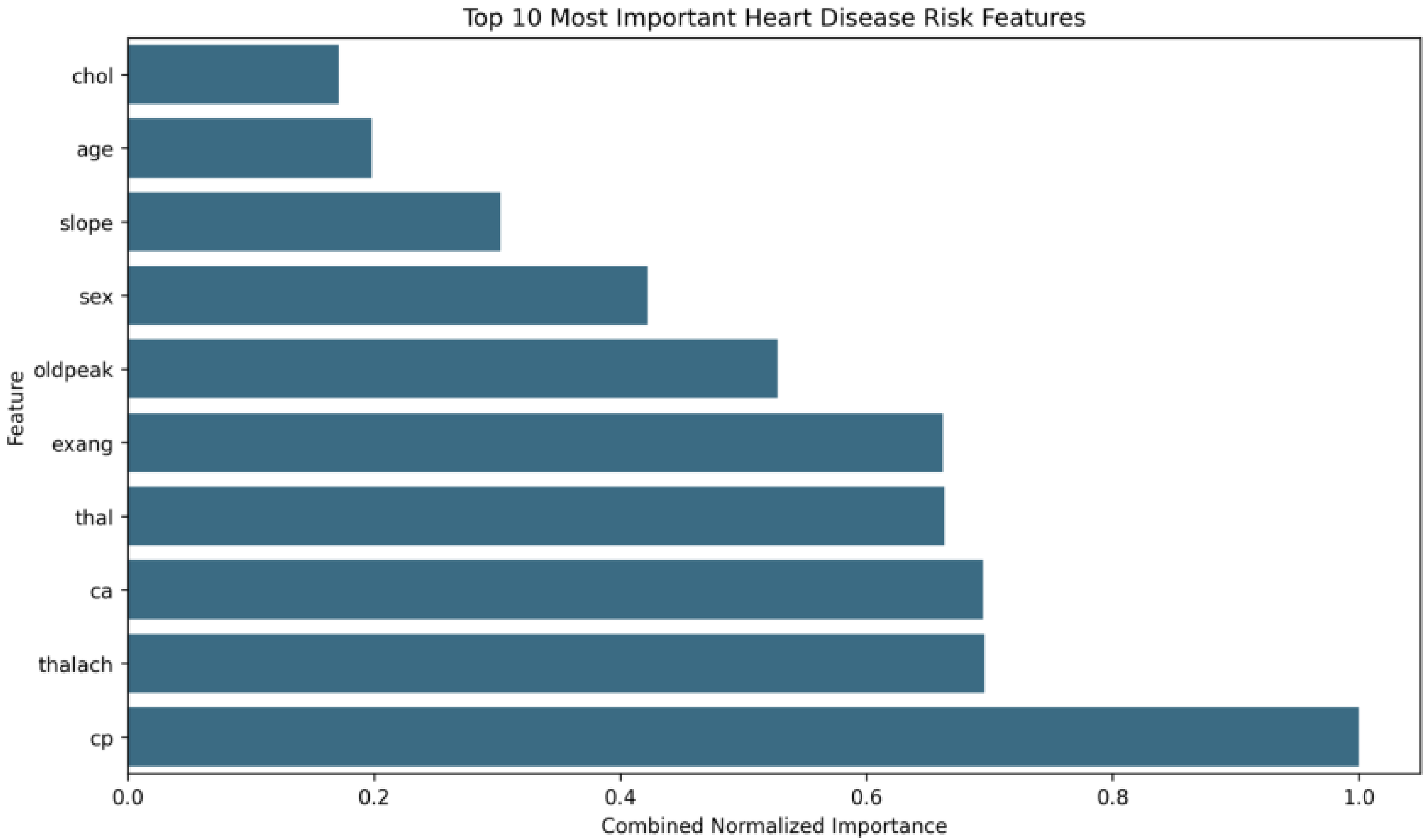
Silhouette Scores



Top 10 Feature Importance

Overall Rank	Feature	Logistic Regression Coefficient	Random Forest Importance	Combined Importance
1.0000	cp	0.2845	0.2102	1.0000
2.0000	thalach	0.2113	0.1364	0.6958
3.0000	ca	-0.2142	0.1338	0.6947
4.0000	thal	-0.1907	0.1379	0.6631
5.0000	exang	-0.2248	0.1121	0.6618
6.0000	oldpeak	-0.1812	0.0880	0.5277
7.0000	sex	-0.1884	0.0381	0.4218
8.0000	slope	0.1269	0.0335	0.3027
9.0000	age	-0.0559	0.0421	0.1984
10.0000	chol	-0.0547	0.0315	0.1712

Feature Importance Chart



Feature Importance Analysis

The leading features in the combined importance ranking were: cp, thalach, ca, thal, exang.

Logistic Regression coefficients provide direction: positive coefficients increase the predicted risk score and negative coefficients decrease it. Random Forest importances rank how strongly each feature contributes to tree split quality, but they do not provide a positive or negative direction.